

The set of near-maximal dual frames

MathlibAnnex.DeterminantFrame.nearMaxFrames

def

Records a determinant deficit while keeping all dual rows contractive.

Statement

Let $n \in \mathbb{N}$, let E be a finite-dimensional real normed vector space, and fix an ordered basis $b = (b_j)_{j=1}^n$. Write $C_b : E \rightarrow \mathbb{R}^n$ for the coordinate isomorphism, defined by $x = \sum_{j=1}^n (C_b x)_j b_j$, and give its coordinate space the sup norm $\|c\|_\infty = \max_j |c_j|$ (zero in dimension zero). A frame is an ordered family $F = (\ell_i)_{i=1}^n$ of continuous real linear functionals, with no independence assumed. Its matrix and evaluation map are

$$M_F = (\ell_i(b_j))_{i,j}, \quad T_F x = (\ell_i(x))_i = M_F C_b x.$$

The admissible frames form the product of closed dual unit balls

$$\mathcal{U} = \{F : \|\ell_i\|_{E^*} \leq 1 \text{ for every } i\}.$$

The norm on E need not be a coordinate sup norm. Write $D_b = \max_{F \in \mathcal{U}} |\det M_F|$. For any $\eta \in \mathbb{R}$, define a set of frames $\mathcal{N}_{b,\eta}$.

Definition

The complete defining set is

$$\mathcal{N}_{b,\eta} = \{F \in \mathcal{U} : D_b - \eta \leq |\det M_F|\}.$$

Assumptions

The space, basis and any real number η are inputs. Neither $\eta \geq 0$ nor $\eta < D_b$ is required to define this set.

Conclusion

A frame belongs precisely when it is admissible and its absolute determinant is at least $D_b - \eta$. This definition neither chooses a frame nor asserts invertibility.

For every real η , compactness of the near-maximal set follows by intersecting compact $\mathcal{U}$ with the closed inequality set. If $\eta \geq 0$, nonemptiness with nonnegative deficit uses an absolute maximizing frame F_* , since $D_b - \eta \leq D_b$. If $\eta < D_b$, membership gives $|\det M_F| \geq D_b - \eta > 0$ and hence invertibility. These are separate consequences with different hypotheses.

Main citations

- compactness of the near-maximal set — `MathlibAnnex.DeterminantFrame.isCompact_nearMaxFrames`
- nonemptiness with nonnegative deficit — `MathlibAnnex.DeterminantFrame.nearMaxFrames_nonempty`
- Exact declaration and proof — `MathlibAnnex.DeterminantFrame.nearMaxFrames`

Lean source signature (exact)

```
-- Frames whose determinant is within `η` of the attained maximum. -/
def nearMaxFrames (b : Basis (Fin n) ℝ E) (η : ℝ) : Set (Frame n E) :=
  {B | B ∈ unitFrameSet n E ∧
    determinantMaximum b - η ≤ |frameDeterminant b B|}
```

The arguments are the basis and the real deficit. The expression following `:=` defines the set.

In the source	Mathematical meaning
<code>n; b : Basis (Fin n) ℝ E</code>	A nonnegative integer n and the ordered basis $(b_j)_{j=1}^n$ of E .
<code>[NormedAddCommGroup E] [NormedSpace ℝ E] [FiniteDimensional ℝ E]</code>	The ambient E is a finite-dimensional real normed vector space, with its given norm (not necessarily a coordinate sup norm). These hypotheses occur in the separate surrounding excerpt.
<code>Fin n → ℝ</code>	The coordinate space $\mathbb{R}^n$ with the sup norm. Lean indexes the coordinates by $0, \dots, n-1$; the formulas here use $1, \dots, n$.
<code>Frame n E</code>	The ordered families $F = (\ell_i)_{i=1}^n$ with $\ell_i : E \rightarrow \mathbb{R}$ continuous and real linear. No independence is part of this type.
<code>unitFrameSet n E</code>	The set $\mathcal{U} = \{F : \ \ell_i\ _{E^*} \leq 1 \text{ for every } i\}$. Its elements are families of functionals, not points of E .
<code>determinantMaximum b</code>	The real number $D_b = \max_{F \in \mathcal{U}} \det(\ell_i(b_j))_{i,j} $, using the specified basis b .
<code>(η : ℝ)</code>	An arbitrary real number η , with no sign assumption.
<code>frameDeterminant b B</code>	$\det M_F$ for the frame F called B , where $(M_F)_{ij} = \ell_i(b_j)$.
In the source	Mathematical meaning
<code>Set (Frame n E)</code>	The output is a set of ordered families of continuous linear functionals.

In the source	Mathematical meaning
$\{B \mid B \in \text{unitFrameSet } n \ E \wedge \text{determinantMaximum } b - \eta \leq \text{frameDeterminant } b \ B \}$	The whole defining set $\mathcal{N}_{b,\eta} = \{F \in \mathcal{U} : D_b - \eta \leq \det M_F \}$. A member must satisfy both conditions.

In $\{B \mid \dots\}$, the bar means “such that”; the bars around `frameDeterminant b B` mean absolute value. $\wedge$ means “and”. Neither condition is an additional requirement for forming the definition.

▼ Relevant surrounding context (separate exact excerpts)

Exact source lines 47–49:

```
variable {n : ℕ}
variable {E : Type u} [NormedAddCommGroup E] [NormedSpace ℝ E]
[FiniteDimensional ℝ E]
```

Full surrounding source. These excerpts are separate from the declaration above.

Surrounding assumptions and aliases: exact source, lines 37–49. The full original context is retained with the source evidence.

▼ Exact content identity

Declaration: `MathlibAnnex.DeterminantFrame.nearMaxFrames`

Accepted content SHA-256: 2e4630db79e0df938f8e4afb4714f26bd877d3aa885d1ca0198f18d1d3320355

Accepted source guide SHA-256: 41ba9ced61ced25e983319b10170466a0d3db2d2c939c300353d611f7b2c90a1

Source commit: 437e6e46228bbb8e91211ebded349d7a30020e73